

Personal Information Management

A local command-line PIM for notes, tasks, events and contacts

COMP3211 · Group 89

Frozen implementation · 4d3508d

Creating a PIR

FR-03 / Task fields

A task needs a description and a local deadline.

COMMAND

```
add task "draft" "2026-10-01 09:00"
```

OBSERVED IN THE FROZEN CLI

```
Created task 2
```

```
id: 2
```

```
type: task
```

```
description: draft
```

```
deadline: 2026-10-01 09:00
```

Defining a search criterion

FR-26 / One search command defines and runs the criterion

```
search (type = "task" && deadline < "2026-10-02 00:00")  
      || text contains "memo"
```

Parentheses group the task and deadline test.
&& requires both conditions; || accepts either branch.

Quoted literals are required. The parser builds criterion objects; it does not run Python input.

Searching for PIRs

FR-27 / Evaluate against the current collection

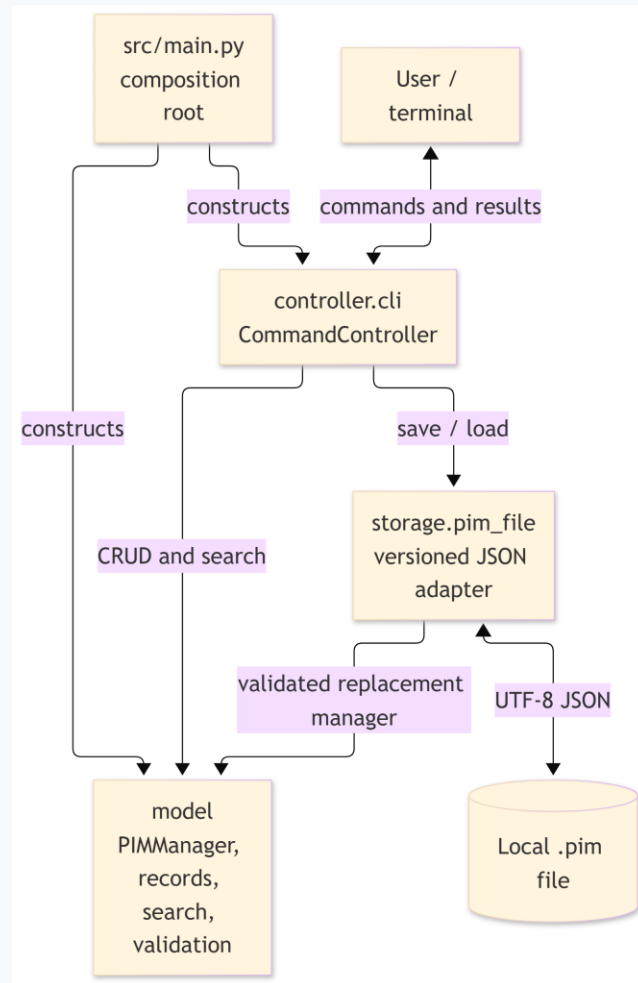
The compound criterion matches two of the four records:

1		note		memo
2		task		draft

FR-39 / Results appear in increasing ID order

After updating task 2 and deleting note 1, a saved snapshot restores three records. The next new note receives ID 5.

Architecture and components



CommandController

Terminal commands and output

PIMManager + model

Records, ID allocation, validation and search

storage.pim_file

Versioned .pim save and fully validated load

The model has no terminal or file I/O.

Lesson from unit testing

FINDING

Deep valid search expressions exceeded Python's recursion limit.

CHANGE

Regression tests led to iterative parsing and evaluation.

WP04 EVIDENCE

33 tests passed · 411/430 model lines covered · 95.58%

Python 3.11 is a target that has not yet been verified.